

# OBS GUI and Config Setting Reference

Generated on 2026-07-15 from dist\OBS\config.ini.

## Searchable Directory

Use PDF search for any GUI label, config key, tab, section, or source file. The companion CSV at docs/settings-reference/settings-index.csv contains the same searchable directory in row form.

## Repository Reference

Private backup/update repository: NANOBYTECOMPUTERS/obs2.

This reference covers the OpenCV 5 CUDA / TensorRT 11.1 project state after neural, depth, PID, feed-forward, and smart-blending code paths were removed. Local dependency drops under modules/ and extras/ are intentionally excluded from Git and should be restored through the documented build/dependency scripts.

## Targeting Pipeline

The runtime path is now detector, tracker, Kalman/IMM state estimation, target-motion state publishing, a paced mouse stream, calibrated pixel-to-count conversion, and mouse output. Detections and tracker observations feed targeting directly without relying on the visual game overlay.

Ego-motion compensation subtracts bounded recent emitted mouse/view motion from tracker priors only. It does not modify raw detections or final mouse output.

## Legacy Prediction Compatibility

Deprecated but retained: minSpeedMultiplier, maxSpeedMultiplier, predictionInterval, snapRadius, nearRadius, speedCurveExponent, snapBoostFactor, kalman\_warmup\_frames, and kalman\_acquisition\_frames remain loadable for old configs and aim-simulation parity.

Removal gate: remove these only after a behavior baseline proves behavior stays identical across the live tracker, game overlay simulation, and config merge path.

| Order | Tab          | Notes                                |
|-------|--------------|--------------------------------------|
| 1     | Capture      | Adjustable controls documented below |
| 2     | Target       | Adjustable controls documented below |
| 3     | Mouse        | Adjustable controls documented below |
| 4     | AI           | Adjustable controls documented below |
| 5     | Buttons      | Adjustable controls documented below |
| 6     | Overlay      | Adjustable controls documented below |
| 7     | Game Overlay | Adjustable controls documented below |
| 8     | Stats        | Read-only monitor tab                |
| 9     | Debug        | Adjustable controls documented below |

# Capture Tab

## Sliders

| Section         | Slider                | Config key            | Range     | Current | Notes                                                                                    |
|-----------------|-----------------------|-----------------------|-----------|---------|------------------------------------------------------------------------------------------|
| General Capture | Capture FPS           | capture_fps           | 0-240     | n/a     | Limits capture loop FPS. 0 disables the cap path and the GUI warns at 0 or 61+.          |
| Capture Preview | Debug scale           | preview_debug_scale   | 0.1x-2.0x | n/a     | Preview-only scale slider. It is not saved to config.ini.                                |
| Virtual Camera  | Virtual camera width  | virtual_camera_width  | 128-3840  | n/a     | Applies when capture_method is virtual_camera.                                           |
| Virtual Camera  | Virtual camera heigth | virtual_camera_heigth | 128-2160  | n/a     | Spelling follows the existing config key. Applies when capture_method is virtual_camera. |

## Activate/Deactivate

| Section         | Control                 | Config key       | Values     | Current | Notes                                                                     |
|-----------------|-------------------------|------------------|------------|---------|---------------------------------------------------------------------------|
| General Capture | Circle mask             | circle_mask      | true/false | n/a     | Applies the circular capture mask.                                        |
| General Capture | Use CUDA Direct Capture | capture_use_cuda | true/false | n/a     | CUDA/TensorRT builds only. Available with duplication_api capture.        |
| Capture Preview | Show Preview Window     | show_window      | true/false | n/a     | Shows the capture preview/debug frame inside the Capture tab.             |
| WinRT           | Capture Borders         | capture_borders  | true/false | n/a     | WinRT capture only. Disabled automatically on unsupported Windows builds. |
| WinRT           | Capture Cursor          | capture_cursor   | true/false | n/a     | WinRT capture only. Includes the cursor in captured frames.               |

## Other GUI-Exposed Config Controls

| Section         | Control                | Config key           | Type/options                                        | Current | Notes                                                                   |
|-----------------|------------------------|----------------------|-----------------------------------------------------|---------|-------------------------------------------------------------------------|
| General Capture | Detection Resolution   | detection_resolution | 160, 320, 640                                       | n/a     | Changes detector input resolution and restarts/reloads dependent paths. |
| General Capture | Capture method         | capture_method       | duplication_api, winrt, virtual_camera, udp_capture | n/a     | Selects frame source.                                                   |
| WinRT           | Capture target (WinRT) | capture_target       | monitor, window                                     | n/a     | WinRT only.                                                             |
| WinRT           | Window title contains  | capture_window_title | text                                                | n/a     | Used when WinRT target is window.                                       |
| Monitor Capture | Capture monitor        | monitor_idx          | monitor index                                       | n/a     | Monitor list is built from active displays.                             |
| Virtual Camera  | Virtual camera         | virtual_camera_name  | available cameras                                   | n/a     | Filtered list of DirectShow video input devices.                        |
| UDP Capture     | UDP IP                 | udp_ip               | IPv4/string                                         | n/a     | Applied with the Apply UDP Settings button.                             |
| UDP Capture     | UDP Port               | udp_port             | 1-65535                                             | n/a     | Applied with the Apply UDP Settings button.                             |

# Target Tab

## Sliders

| Section | Slider                    | Config key    | Range     | Current | Notes                                              |
|---------|---------------------------|---------------|-----------|---------|----------------------------------------------------|
| Offsets | Approximate Body Y Offset | body_y_offset | 0.05-0.90 | n/a     | Vertical aim point used for body-class detections. |
| Offsets | Approximate Head Y Offset | head_y_offset | 0.05-0.55 | n/a     | Vertical aim point used for head-class detections. |

## Activate/Deactivate

| Section   | Control          | Config key       | Values     | Current | Notes                                                                                      |
|-----------|------------------|------------------|------------|---------|--------------------------------------------------------------------------------------------|
| Targeting | Disable Headshot | disable_headshot | true/false | n/a     | Prevents the head class from being selected as the aim target.                             |
| Targeting | Auto Aim         | auto_aim         | true/false | n/a     | Allows target selection/movement logic to act automatically while held conditions are met. |

# Mouse Tab

## Sliders

| Section                   | Slider                      | Config key                           | Range            | Current | Notes                                                                                                    |
|---------------------------|-----------------------------|--------------------------------------|------------------|---------|----------------------------------------------------------------------------------------------------------|
| State Estimator           | Acquisition frames          | kalman_acquisition_frames            | 3-5              | n/a     | Frames used to ramp prediction weight after target acquisition.                                          |
| State Estimator           | Process noise position      | kalman_process_noise_position        | 0.0001-5000      | n/a     | Position process noise. Higher values adapt faster to position changes.                                  |
| State Estimator           | Process noise velocity      | kalman_process_noise_velocity        | 0.0001-50000     | n/a     | Velocity process noise. Higher values adapt faster to speed changes.                                     |
| State Estimator           | Measurement noise           | kalman_measurement_noise             | 0.0001-5000      | n/a     | Measurement noise. Higher values trust detections less.                                                  |
| State Estimator           | Velocity damping            | kalman_velocity_damping              | 0.0-3.0          | n/a     | Damps velocity estimates to reduce runaway prediction.                                                   |
| State Estimator           | Max velocity (px/s)         | kalman_max_velocity                  | 100-60000        | n/a     | Maximum allowed tracked velocity.                                                                        |
| State Estimator           | Warmup frames               | kalman_warmup_frames                 | 0-20             | n/a     | Legacy warmup frames when latency sweep is off.                                                          |
| State Estimator           | Additional prediction (ms)  | kalman_additional_prediction_ms      | -80.0-120.0      | n/a     | Manual extra prediction offset in milliseconds.                                                          |
| State Estimator           | Reset timeout (s)           | kalman_reset_timeout_sec             | 0.05-3.0         | n/a     | Time without stable observations before estimator state resets.                                          |
| State Estimator           | Ego compensation strength   | ego_motion_compensation_strength     | 0.0-1.0          | n/a     | Fraction of recent emitted motion subtracted from tracker priors/history to reduce camera-motion jitter. |
| State Estimator           | Ego max shift (px @640)     | ego_motion_compensation_max_shift_px | 1-128            | n/a     | Per-frame compensation clamp at 640 detection resolution; scales with detection_resolution.              |
| State Estimator           | Ego max age (ms)            | ego_motion_compensation_max_age_ms   | 16-500           | n/a     | Drops stale emitted-motion samples so compensation fails closed.                                         |
| Direct Targeting Movement | Deadzone (px)               | target_deadzone_px                   | 0.000-20.000     | n/a     | Minimum target error radius before stream movement is emitted.                                           |
| Direct Targeting Movement | Stream interval (ms)        | target_stream_interval_ms            | 0.50-8.00        | n/a     | Paced target-stream wake interval. New tracker observations wake the stream immediately.                 |
| Direct Targeting Movement | Stream sharpness            | target_stream_sharpness              | 1.00-80.00       | n/a     | Time-based convergence factor for the target stream.                                                     |
| Direct Targeting Movement | Max speed (px/s)            | target_max_pixel_speed               | 50-20000         | n/a     | Maximum pixel-space stream speed before count conversion.                                                |
| Direct Targeting Movement | State max age (ms)          | target_state_max_age_ms              | 16-500           | n/a     | Drops stale target-motion snapshots before streaming output.                                             |
| Direct Targeting Movement | Min stream confidence       | target_min_stream_confidence         | 0.00-0.95        | n/a     | Minimum confidence required for the target stream to emit movement.                                      |
| Direct Targeting Movement | Prediction blend            | target_prediction_blend              | 0.000-0.650      | n/a     | Fraction of bounded short-horizon predicted aim mixed into the observed tracker aim point.               |
| Direct Targeting Movement | Max prediction lead (px)    | target_prediction_max_lead_px        | 0.00-40.00       | n/a     | Maximum pixel distance the mouse path may lead the current tracker aim point.                            |
| Direct Targeting Movement | Fallback max step (px/call) | target_max_pixel_step                | 0.25-120.00      | n/a     | Legacy per-call clamp for non-stream direct movement helpers and aim-simulation parity.                  |
| Direct Targeting Movement | Fallback output scale       | target_output_scale                  | 0.010-3.000      | n/a     | Legacy per-call scale for non-stream direct movement helpers and aim-simulation parity.                  |
| Direct Targeting Movement | Counts / px X               | target_counts_per_pixel_x            | -50.0000-50.0000 | n/a     | Horizontal calibrated mouse counts per screen-space pixel.                                               |
| Direct Targeting Movement | Counts / px Y               | target_counts_per_pixel_y            | -50.0000-50.0000 | n/a     | Vertical calibrated mouse counts per screen-space pixel.                                                 |
| Auto Shoot                | bScope Multiplier           | bScope_multiplier                    | 0.5-2.0          | n/a     | Multiplier used by auto-shoot scope timing/behavior. Disabled in UI until Auto Shoot is on.              |

## Activate/Deactivate

| Section                   | Control                    | Config key                        | Values     | Current | Notes                                                                                                                  |
|---------------------------|----------------------------|-----------------------------------|------------|---------|------------------------------------------------------------------------------------------------------------------------|
| State Estimator           | Runtime latency sweep      | runtime_latency_sweep_enabled     | true/false | n/a     | Enables side-by-side latency sweep behavior for estimator and PID experiments.                                         |
| State Estimator           | Enable Kalman estimator    | kalman_enabled                    | true/false | n/a     | Enables Kalman filtering for target motion estimation.                                                                 |
| State Estimator           | Seed velocity on acquire   | kalman_velocity_seed_enabled      | true/false | n/a     | Seeds velocity from early measurement deltas during acquisition.                                                       |
| State Estimator           | Compensate detection delay | kalman_compensate_detection_delay | true/false | n/a     | Accounts for detector latency in prediction.                                                                           |
| State Estimator           | Ego-motion compensation    | ego_motion_compensation_enabled   | true/false | n/a     | Opt-in tracker-prior compensation from emitted mouse/view motion. Raw detections and final mouse output are unchanged. |
| Direct Targeting Movement | Target stream              | target_stream_enabled             | true/false | n/a     | Enables the paced target-motion stream used by the live targeting path.                                                |

| Section                   | Control                 | Config key                             | Values     | Current | Notes                                                                                                             |
|---------------------------|-------------------------|----------------------------------------|------------|---------|-------------------------------------------------------------------------------------------------------------------|
| Direct Targeting Movement | Target stream debug     | target_stream_debug_enabled            | true/false | n/a     | Shows live target-stream status, prediction, pixel delta, count output, and block reason in the Mouse tab.        |
| Direct Targeting Movement | Calibrated pixel counts | target_calibrated_pixel_counts_enabled | true/false | n/a     | Uses measured counts-per-pixel gains for direct targeting output instead of the default 1:1 pixel-count fallback. |
| Auto Shoot                | Auto Shoot              | auto_shoot                             | true/false | n/a     | Enables automatic shooting behavior.                                                                              |
| Input Method              | Arduino 16-bit Mouse    | arduino_16_bit_mouse                   | true/false | n/a     | Arduino input only. Sends wider mouse movement values.                                                            |
| Input Method              | Arduino Enable Keys     | arduino_enable_keys                    | true/false | n/a     | Arduino input only. Enables keyboard key output through Arduino.                                                  |

## Other GUI-Exposed Config Controls

| Section      | Control                 | Config key       | Type/options                                                                           | Current | Notes                                |
|--------------|-------------------------|------------------|----------------------------------------------------------------------------------------|---------|--------------------------------------|
| Input Method | Mouse Input Method      | input_method     | WIN32, GHUB, RAZER, DIRECT, ARDUINO, TEENSY41, TEENSY41_HID, KMBOX_NET, KMBOX_A, MAKCU | n/a     | Changes active mouse backend.        |
| Input Method | Arduino/Teensy Port     | arduino_port     | COM1-COM30                                                                             | n/a     | Arduino and Teensy 4.1 serial input. |
| Input Method | Arduino/Teensy Baudrate | arduino_baudrate | 9600-115200                                                                            | n/a     | Arduino and Teensy 4.1 serial input. |
| Input Method | Kmbox Net IP            | kmbox_net_ip     | text                                                                                   | n/a     | Saved with Save & Reconnect.         |
| Input Method | Kmbox Net Port          | kmbox_net_port   | text                                                                                   | n/a     | Saved with Save & Reconnect.         |
| Input Method | Kmbox Net UUID          | kmbox_net_uuid   | text                                                                                   | n/a     | Saved with Save & Reconnect.         |
| Input Method | Kmbox A PIDVID          | kmbox_a_pidvid   | PPPPVVVV                                                                               | n/a     | Saved with Save & Reconnect.         |
| Input Method | Makcu Port              | makcu_port       | AUTO, COM1-COM30                                                                       | n/a     | Makcu input only.                    |
| Input Method | Makcu Baudrate          | makcu_baudrate   | 115200-4000000                                                                         | n/a     | Makcu input only.                    |

# AI Tab

## Sliders

| Section   | Slider               | Config key           | Range     | Current | Notes                                                      |
|-----------|----------------------|----------------------|-----------|---------|------------------------------------------------------------|
| Detection | Confidence Threshold | confidence_threshold | 0.01-1.00 | n/a     | Minimum detector confidence accepted as a detection.       |
| Detection | NMS Threshold        | nms_threshold        | 0.00-1.00 | n/a     | Non-max suppression overlap threshold.                     |
| Detection | Max Detections       | max_detections       | 1-100     | n/a     | Maximum detections retained after model output processing. |

## Activate/Deactivate

No activate/deactivate checkboxes in this tab.

## Other GUI-Exposed Config Controls

| Section | Control | Config key | Type/options  | Current | Notes                                                   |
|---------|---------|------------|---------------|---------|---------------------------------------------------------|
| Model   | Model   | ai_model   | models folder | n/a     | Selects detector model from available files.            |
| Backend | Backend | backend    | TRT, DML      | n/a     | CUDA builds only; DML build does not expose this combo. |

# Buttons Tab

## Sliders

No saved GUI sliders in this tab.

## Activate/Deactivate

| Section           | Control                    | Config key             | Values     | Current | Notes                                          |
|-------------------|----------------------------|------------------------|------------|---------|------------------------------------------------|
| Arrow Key Options | Enable arrows keys options | enable_arrows_settings | true/false | n/a     | Allows arrow-key adjustment of target offsets. |

## Other GUI-Exposed Config Controls

| Section         | Control               | Config key           | Type/options | Current | Notes                                    |
|-----------------|-----------------------|----------------------|--------------|---------|------------------------------------------|
| Button Bindings | Targeting Buttons     | button_targeting     | key list     | n/a     | Supports multiple bindings with + and -. |
| Button Bindings | Shoot Buttons         | button_shoot         | key list     | n/a     | Supports multiple bindings with + and -. |
| Button Bindings | Zoom Buttons          | button_zoom          | key list     | n/a     | Supports multiple bindings with + and -. |
| Button Bindings | Exit Buttons          | button_exit          | key list     | n/a     | Supports multiple bindings with + and -. |
| Button Bindings | Pause Buttons         | button_pause         | key list     | n/a     | Supports multiple bindings with + and -. |
| Button Bindings | Reload Config Buttons | button_reload_config | key list     | n/a     | Supports multiple bindings with + and -. |
| Button Bindings | Overlay Buttons       | button_open_overlay  | key list     | n/a     | Supports multiple bindings with + and -. |

# Overlay Tab

## Sliders

| Section | Slider          | Config key       | Range     | Current | Notes                                                       |
|---------|-----------------|------------------|-----------|---------|-------------------------------------------------------------|
| Visual  | Overlay Opacity | overlay_opacity  | 220-255   | n/a     | Window opacity. The UI clamps to a readable minimum of 220. |
| Visual  | UI Fine Scale   | overlay_ui_scale | 0.85-1.35 | n/a     | Fine-grained scaling for the overlay UI.                    |

## Activate/Deactivate

| Section         | Control                      | Config key                   | Values     | Current | Notes                                                                                           |
|-----------------|------------------------------|------------------------------|------------|---------|-------------------------------------------------------------------------------------------------|
| Capture Privacy | Hide Overlays From Recording | overlay_exclude_from_capture | true/false | n/a     | Applies capture exclusion so overlay windows are hidden from supported recording/capture paths. |

## Other GUI-Exposed Config Controls

| Section | Control     | Config key | Type/options      | Current | Notes                                                                                |
|---------|-------------|------------|-------------------|---------|--------------------------------------------------------------------------------------|
| Config  | Save Config | config.ini | writes config.ini | n/a     | Immediately writes the current runtime config.                                       |
| Config  | Load Config | config.ini | reads config.ini  | n/a     | Merges config.ini into current runtime values and refreshes runtime-sensitive paths. |



# Game Overlay Tab

## Sliders

| Section            | Slider                         | Config key                        | Range        | Current | Notes                                                            |
|--------------------|--------------------------------|-----------------------------------|--------------|---------|------------------------------------------------------------------|
| General            | Overlay Max FPS (0 = uncapped) | game_overlay_max_fps              | 0-256        | n/a     | Caps game overlay render rate. 0 is uncapped.                    |
| Box Color          | A                              | game_overlay_box_a                | 0-255        | n/a     | Detection box alpha.                                             |
| Box Color          | R                              | game_overlay_box_r                | 0-255        | n/a     | Detection box red channel.                                       |
| Box Color          | G                              | game_overlay_box_g                | 0-255        | n/a     | Detection box green channel.                                     |
| Box Color          | B                              | game_overlay_box_b                | 0-255        | n/a     | Detection box blue channel.                                      |
| Box Color          | Box Thickness                  | game_overlay_box_thickness        | 0.5-10.0     | n/a     | Detection box line thickness.                                    |
| Capture Frame      | A                              | game_overlay_frame_a              | 0-255        | n/a     | Capture frame alpha.                                             |
| Capture Frame      | R                              | game_overlay_frame_r              | 0-255        | n/a     | Capture frame red channel.                                       |
| Capture Frame      | G                              | game_overlay_frame_g              | 0-255        | n/a     | Capture frame green channel.                                     |
| Capture Frame      | B                              | game_overlay_frame_b              | 0-255        | n/a     | Capture frame blue channel.                                      |
| Capture Frame      | Frame Thickness                | game_overlay_frame_thickness      | 0.5-10.0     | n/a     | Capture frame line thickness.                                    |
| Future Point Style | Point Radius                   | game_overlay_future_point_radius  | 1.0-20.0     | n/a     | Radius of future-position points.                                |
| Future Point Style | Point Step Alpha Falloff       | game_overlay_future_alpha_falloff | 0.10-5.00    | n/a     | Controls how quickly future-point alpha fades along the path.    |
| Icon Overlay       | Icon Width                     | game_overlay_icon_width           | 4-512        | n/a     | Icon width in pixels.                                            |
| Icon Overlay       | Icon Height                    | game_overlay_icon_height          | 4-512        | n/a     | Icon height in pixels.                                           |
| Icon Overlay       | Icon Offset X                  | game_overlay_icon_offset_x        | -500.0-500.0 | n/a     | Horizontal icon offset from the selected anchor.                 |
| Icon Overlay       | Icon Offset Y                  | game_overlay_icon_offset_y        | -500.0-500.0 | n/a     | Vertical icon offset from the selected anchor.                   |
| Aim Simulation     | Sim X                          | aim_sim_x                         | -3000-3000   | n/a     | Aim simulation window X position.                                |
| Aim Simulation     | Sim Y                          | aim_sim_y                         | -3000-3000   | n/a     | Aim simulation window Y position.                                |
| Aim Simulation     | Sim Width                      | aim_sim_width                     | 220-1600     | n/a     | Aim simulation window width.                                     |
| Aim Simulation     | Sim Height                     | aim_sim_height                    | 180-1000     | n/a     | Aim simulation window height.                                    |
| Aim Simulation     | Sim FPS Min                    | aim_sim_fps_min                   | 15-360       | n/a     | Minimum simulated frame rate.                                    |
| Aim Simulation     | Sim FPS Max                    | aim_sim_fps_max                   | 15-360       | n/a     | Maximum simulated frame rate. The GUI keeps min and max ordered. |
| Aim Simulation     | FPS Jitter                     | aim_sim_fps_jitter                | 0.000-0.800  | n/a     | Randomized FPS variance for the simulation.                      |
| Aim Simulation     | Capture Delay (ms)             | aim_sim_capture_delay_ms          | 0.0-80.0     | n/a     | Simulated capture delay.                                         |
| Aim Simulation     | Inference Delay (ms)           | aim_sim_inference_delay_ms        | 0.0-120.0    | n/a     | Manual simulated inference delay when not using live timing.     |
| Aim Simulation     | Input Delay (ms)               | aim_sim_input_delay_ms            | 0.0-60.0     | n/a     | Simulated input delay.                                           |
| Aim Simulation     | Extra Delay (ms)               | aim_sim_extra_delay_ms            | 0.0-60.0     | n/a     | Additional simulated processing delay.                           |
| Aim Simulation     | Target Max Speed               | aim_sim_target_max_speed          | 20-2500      | n/a     | Maximum simulated target speed.                                  |
| Aim Simulation     | Target Accel                   | aim_sim_target_accel              | 20-10000     | n/a     | Simulated target acceleration.                                   |
| Aim Simulation     | Target Stop Chance             | aim_sim_target_stop_chance        | 0.00-0.95    | n/a     | Probability that the simulated target pauses during a retarget.  |

## Activate/Deactivate

| Section       | Control                | Config key                          | Values     | Current | Notes                                                             |
|---------------|------------------------|-------------------------------------|------------|---------|-------------------------------------------------------------------|
| General       | Enable                 | game_overlay_enabled                | true/false | n/a     | Enables the separate game overlay renderer.                       |
| General       | Draw Detection Boxes   | game_overlay_draw_boxes             | true/false | n/a     | Shows detection boxes in the game overlay.                        |
| General       | Draw Future Positions  | game_overlay_draw_future            | true/false | n/a     | Shows predicted future target points.                             |
| General       | Draw Wind Debug Tail   | game_overlay_draw_wind_tail         | true/false | n/a     | Shows the wind-mouse debug trail when wind motion is active.      |
| General       | Show Target Correction | game_overlay_show_target_correction | true/false | n/a     | Shows correction indicators for target prediction/aim adjustment. |
| Capture Frame | Draw Capture Frame     | game_overlay_draw_frame             | true/false | n/a     | Draws the capture frame rectangle.                                |
| Icon Overlay  | Enable Icon Overlay    | game_overlay_icon_enabled           | true/false | n/a     | Enables drawing an icon relative to detections.                   |

| Section        | Control                      | Config key                 | Values     | Current | Notes                                                                     |
|----------------|------------------------------|----------------------------|------------|---------|---------------------------------------------------------------------------|
| Aim Simulation | Enable Aim Simulation Window | aim_sim_enabled            | true/false | n/a     | Shows the aim simulation overlay window.                                  |
| Aim Simulation | Use Live Inference Delay     | aim_sim_use_live_inference | true/false | n/a     | Uses live backend timing for inference delay instead of the manual value. |
| Aim Simulation | Show Delayed Observation     | aim_sim_show_observed      | true/false | n/a     | Shows delayed target observation in the simulation.                       |
| Aim Simulation | Show Trajectory History      | aim_sim_show_history       | true/false | n/a     | Shows historical target/aim trails.                                       |
| Aim Simulation | Show Kalman Debug            | aim_sim_show_kalman_debug  | true/false | n/a     | Shows Kalman-related debug visuals in the simulation.                     |

### Other GUI-Exposed Config Controls

| Section      | Control               | Config key               | Type/options              | Current | Notes                                  |
|--------------|-----------------------|--------------------------|---------------------------|---------|----------------------------------------|
| Icon Overlay | Icon Path             | game_overlay_icon_path   | path                      | n/a     | Image path for icon overlay.           |
| Icon Overlay | Icon Class (-1 = all) | game_overlay_icon_class  | -1 or class id            | n/a     | Restricts icon to one class when >= 0. |
| Icon Overlay | Icon Anchor           | game_overlay_icon_anchor | center, top, bottom, head | n/a     | Anchor point used for icon placement.  |

## Stats Tab

The Stats tab is read-only in the current GUI. It displays timing graphs, capture FPS, capture details, and CUDA status where available.

# Debug Tab

## Sliders

No saved GUI sliders in this tab.

## Activate/Deactivate

| Section            | Control                | Config key             | Values     | Current | Notes                                            |
|--------------------|------------------------|------------------------|------------|---------|--------------------------------------------------|
| Screenshot Buttons | Verbose console output | verbose                | true/false | n/a     | Enables extra console logging.                   |
| Log File           | Enable log file        | debug_log_file_enabled | true/false | n/a     | Writes logs to debug_log_file_path when enabled. |

## Other GUI-Exposed Config Controls

| Section            | Control            | Config key          | Type/options | Current | Notes                                    |
|--------------------|--------------------|---------------------|--------------|---------|------------------------------------------|
| Screenshot Buttons | Screenshot buttons | screenshot_button   | key list     | n/a     | Supports multiple bindings with + and -. |
| Screenshot Buttons | Screenshot delay   | screenshot_delay    | step 50/500  | n/a     | Delay before screenshot capture.         |
| Log File           | Log file path      | debug_log_file_path | path         | n/a     | Used when Enable log file is on.         |

# Config.ini Settings Not In The GUI

These settings are editable by changing config.ini directly. Some are build-gated: CUDA rows are emitted only by CUDA builds.

| Section          | Setting                    | Current/default | Range/options      | Visibility                 | Guidance                                                                      | Notes                                                                              |
|------------------|----------------------------|-----------------|--------------------|----------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Mouse prediction | prediction_futurePositions | 20              | integer            | Config only                | Number of future target positions to retain/use for prediction visualization. | Saved in current config.ini, but no direct GUI control exists.                     |
| Mouse prediction | draw_futurePositions       | true            | true/false         | Config only                | Enables drawing future positions in preview/debug paths.                      | Separate from game_overlay_draw_future.                                            |
| AI backend       | backend                    | DML             | DML or TRT         | DML config-only / CUDA GUI | Selects DirectML or TensorRT backend.                                         | GUI combo exists only in CUDA builds.                                              |
| AI backend       | dml_device_id              | 0               | integer adapter id | Config only                | DirectML adapter index used by ONNX Runtime DML.                              | Not exposed in the GUI.                                                            |
| System reserves  | cpuCoreReserveCount        | 4               | integer            | Config only                | CPU cores reserved away from worker assignment.                               | Not exposed in the GUI.                                                            |
| System reserves  | systemMemoryReserveMB      | 2048            | MB                 | Config only                | System memory reserve used by runtime resource planning.                      | Not exposed in the GUI.                                                            |
| Custom classes   | class_player               | 0               | integer class id   | Config only                | Detector class id treated as player/body.                                     | Not exposed in the GUI.                                                            |
| Custom classes   | class_head                 | 1               | integer class id   | Config only                | Detector class id treated as head.                                            | Not exposed in the GUI.                                                            |
| Debug            | show_fps                   | false           | true/false         | Config only                | Legacy/debug FPS display flag.                                                | Saved in config.ini but not exposed in the current GUI.                            |
| Advanced mouse   | minSpeedMultiplier         | 0.1             | float              | Loadable hidden key        | Legacy minimum movement speed multiplier.                                     | Loadable from config.ini if added manually; saveConfig does not currently emit it. |
| Advanced mouse   | maxSpeedMultiplier         | 0.1             | float              | Loadable hidden key        | Legacy maximum movement speed multiplier.                                     | Loadable from config.ini if added manually; saveConfig does not currently emit it. |
| Advanced mouse   | predictionInterval         | 0.01            | seconds            | Loadable hidden key        | Prediction sample interval used by legacy/auxiliary prediction logic.         | Loadable from config.ini if added manually; saveConfig does not currently emit it. |
| Legacy snap      | snapRadius                 | 1.5             | float              | Config only                | Legacy snap radius.                                                           | Loadable hidden key; no GUI control.                                               |
| Legacy snap      | nearRadius                 | 25.0            | float              | Config only                | Legacy near-target radius.                                                    | Loadable hidden key; no GUI control.                                               |
| Legacy snap      | speedCurveExponent         | 3.0             | float              | Config only                | Legacy speed curve exponent.                                                  | Loadable hidden key; no GUI control.                                               |
| Legacy snap      | snapBoostFactor            | 1.15            | float              | Config only                | Legacy snap boost multiplier.                                                 | Loadable hidden key; no GUI control.                                               |
| Legacy recoil    | easynorecoil               | false           | true/false         | Config only                | Legacy recoil helper flag.                                                    | Loadable hidden key; no GUI control and not emitted by saveConfig.                 |
| Legacy recoil    | easynorecoilstrength       | 0.0             | float              | Config only                | Legacy recoil helper strength.                                                | Loadable hidden key; no GUI control and not emitted by saveConfig.                 |
| Wind mouse       | wind_mouse_enabled         | false           | true/false         | Config only                | Enables wind-style motion noise in supported runtime/simulation paths.        | Loadable hidden key; no GUI control.                                               |
| Wind mouse       | wind_G                     | 18.0            | float              | Config only                | Wind mouse gravity parameter.                                                 | Loadable hidden key; no GUI control.                                               |
| Wind mouse       | wind_W                     | 15.0            | float              | Config only                | Wind mouse wind parameter.                                                    | Loadable hidden key; no GUI control.                                               |
| Wind mouse       | wind_M                     | 10.0            | float              | Config only                | Wind mouse max-step parameter.                                                | Loadable hidden key; no GUI control.                                               |
| Wind mouse       | wind_D                     | 8.0             | float              | Config only                | Wind mouse damping/distance parameter.                                        | Loadable hidden key; no GUI control.                                               |
| CUDA             | export_enable_fp8          | false           | true/false         | CUDA config-only           | Enables FP8 TensorRT export where supported.                                  | CUDA builds only; no GUI checkbox.                                                 |
| CUDA             | export_enable_fp16         | true            | true/false         | CUDA config-only           | Enables FP16 TensorRT export where supported.                                 | CUDA builds only; no GUI checkbox.                                                 |
| CUDA             | use_cuda_graph             | false           | true/false         | CUDA config-only           | Enables CUDA graph execution path.                                            | CUDA builds only; no GUI checkbox.                                                 |
| CUDA             | use_pinned_memory          | true            | true/false         | CUDA config-only           | Uses pinned host memory for CUDA paths.                                       | CUDA builds only; no GUI checkbox.                                                 |
| CUDA             | gpuMemoryReserveMB         | 2048            | MB                 | CUDA config-only           | GPU memory reserve for CUDA resource planning.                                | CUDA builds only; no GUI control.                                                  |
| CUDA             | enableGpuExclusiveMode     | true            | true/false         | CUDA config-only           | Attempts GPU exclusive mode at startup.                                       | CUDA builds only; no GUI checkbox.                                                 |

# Source Map

| Area                      | Source                                              |
|---------------------------|-----------------------------------------------------|
| Tab order                 | overlay/overlay.cpp                                 |
| Capture                   | overlay/draw_capture.cpp and overlay/draw_debug.cpp |
| Target                    | overlay/draw_target.cpp                             |
| Mouse                     | overlay/draw_mouse.cpp                              |
| Buttons                   | overlay/draw_buttons.cpp                            |
| Overlay                   | overlay/draw_overlay.cpp                            |
| Game Overlay              | overlay/draw_game_overlay.cpp                       |
| Debug                     | overlay/draw_debug.cpp                              |
| Config load/save/defaults | config/config.cpp and config/config.h               |